

[illegible]

2 - 4-SWITCH BUCK-BOOST (MCU PWM – no switcher IC)

The diagram illustrates a 4-switch buck-boost converter circuit. It features two MOSFETs (Q1, Q2, Q3, Q4) and two MOSFET drivers (U2, U3). The input is VBUS and the output is VPRE. The MOSFETs are AON6324, the inductor is 22uH 5A, the diodes are 1N4148W, and the capacitors are 22uF/35V. The MOSFET drivers are IR2104. The circuit is controlled by MCU PWM signals (PWM_A, EN_A, PWM_B, EN_B) and gate drive signals (G_A_HI, G_A_LO, G_B_HI, G_B_LO).

[illegible]

4 - +5 V LDO

U5
LP2950-5.0

The diagram shows the +5V LDO regulator (U5, LP2950-5.0) connected between the VBUS and GND. The input pin (VI) is connected to VBUS, and the output pin (VO) is connected to the +5V output. The GND pin (2) is connected to the common ground. Two capacitors, C6 (10uF) and C7 (10uF), are connected to the input and output pins respectively.

6 – PROG / UART

J3
PROG



The diagram shows a 4-pin connector labeled J3. The pins are numbered 1, 2, 3, and 4 from top to bottom. Pin 1 is connected to SWDIO, Pin 2 is connected to SWCLK, and Pin 3 is connected to UART_TX. Pin 4 is connected to ground, indicated by a triangle symbol.

5 - MCU

U6
CH32V203C8T5

C8
100nF

C9
100nF

R9
10k

R10
10k

Buttons: BTN_SEL, BTN_OUT, BTN_BACK

LEDs: LED_HB, LED_CC

UART: UART_TX, UART_RX

I2C: I2C_SDA, I2C_SCL

ENC: ENC_A, ENC_B, ENC_SW

MCU Pins: RST, BOOT0, OSC_IN, OSC_OUT, PC13, PC14, PC15, PB0, PB1, PB2, PB3, PB4, PB5, PB6, PB7, PB8, PB9, PB10, PB11, PB12, PB13, PB14, PB15, PA0, PA1, PA2, PA3, PA4, PA5, PA6, PA7, PA8, PA9, PA10, PA11, PA12, SWDIO, SWCLK, PA15

ADC: ADC_VBUS, ADC_VPRE, ADC_ISNS, PWM_REF

PWM: PWM_A, PWM_B

EN: EN_A, EN_B

SW: SWDIO, SWCLK, LED_OUT

Power: VBAT, VDD, VDDA, VSS, VSSA

No switcher IC - Dead-time and b

7 - USER INTERFACE

J4
OLED 0.96in I2C

1
2
3
4

I2C_SCL
I2C_SDA

SW1
EC11

ENC_A
ENC_B

A
C
B

S1
S2

ENC_SW

SW2
OUTPUT

SW3
SELECT

SW4
BACK

BTN_OUT
BTN_SEL
BTN_BACK

D4
red

D5
amber

D6
green

R11
1k

R12
1k

R13
1k

LED-OUT
LED-CC
LED-HB

MCU drives all four gates. BUCK: leg A chops. Q4 parked on. BOOST: Q1 parked on, leg B chops. Trap refresh in firmware.

The diagram illustrates the user interface components and their connections. At the top left, an I2C OLED display (J4, 0.96in) is connected to the I2C_SCL and I2C_SDA pins. Below it, a push button (SW1, EC11) is connected to the MCU. To the right, three push buttons (SW2, SW3, SW4) are connected to the MCU via buttons (BTN_OUT, BTN_SEL, BTN_BACK). At the bottom left, an encoder (ENC_A, ENC_B, A, B, C, S1, S2) is connected to the MCU. At the bottom right, three LEDs (D4 red, D5 amber, D6 green) are connected to the MCU via resistors (R11, R12, R13). A text box explains that the MCU drives all four gates, with BUCK leg A chopping, Q4 parked on, and BOOST leg B chopping, with Q1 parked on. It also mentions a trap refresh in the firmware.

8 – HARDWARE OVER-CURRENT TRIP

The diagram illustrates a hardware over-current trip circuit. It consists of two LM2903 comparators, U7A and U7B. U7A's non-inverting input (+) is connected to a voltage divider (R14, 10k 1%) from VCC. Its inverting input (-) is connected to the ADC_IN5 pin (R15, 4k7 1%) and the output of U7B. U7B's non-inverting input (+) is connected to the output of U7A. Its inverting input (-) is connected to the EN_A pin. The outputs of both comparators are connected to a common output line. A green arrow indicates the signal path from the input divider through the comparators to the output.

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